

Concavity & Inflection Points

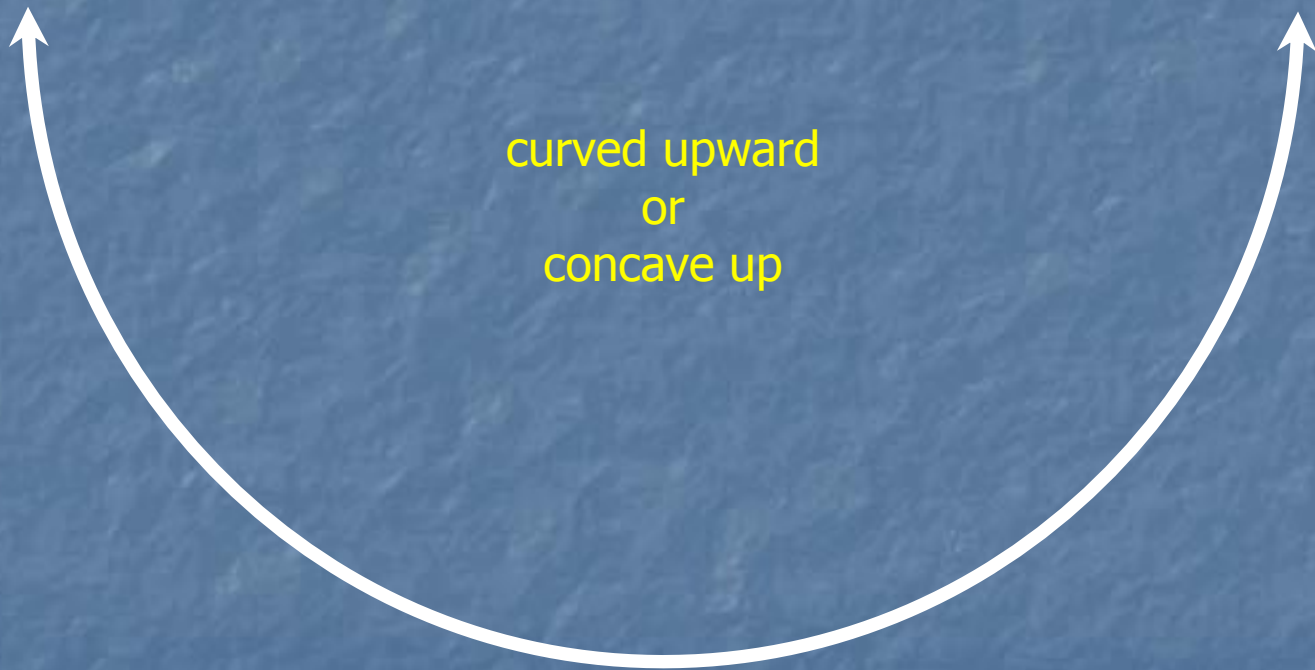
Objectives

- To determine the intervals on which the graph of a function is concave up or concave down.
- To find the inflection points of a graph of a function.

Concavity

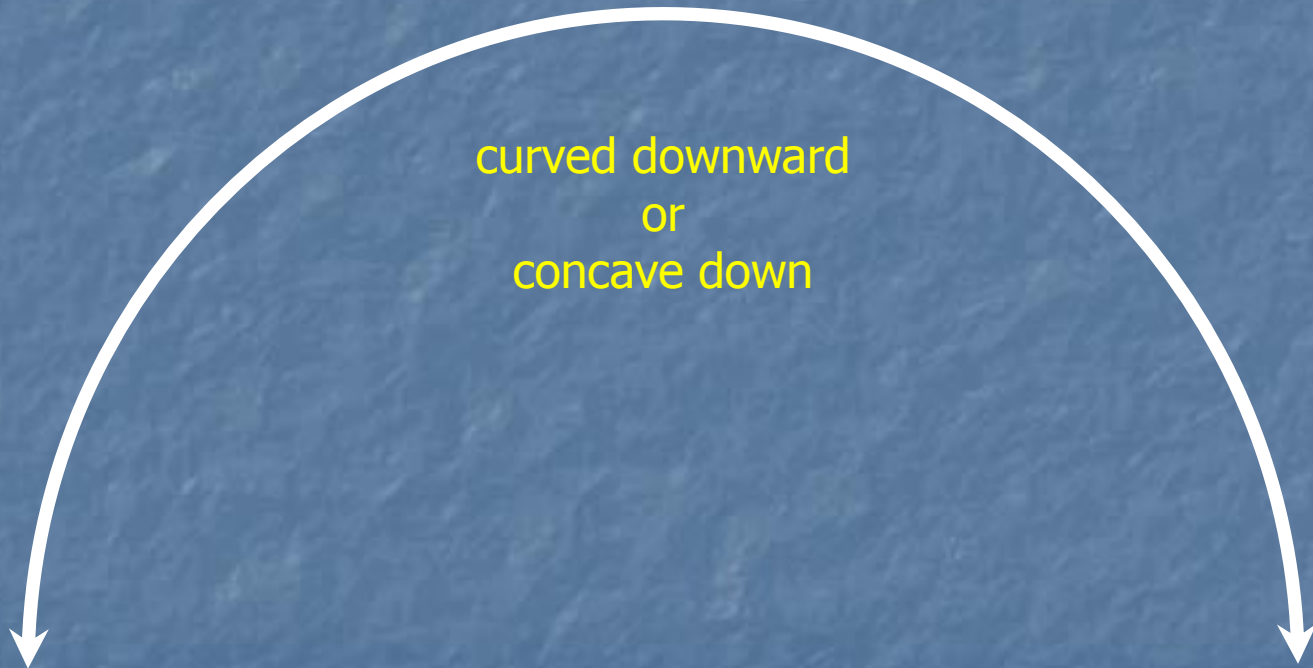
- The **concavity** of the graph of a function is the notion of curving upward or downward.

Concavity



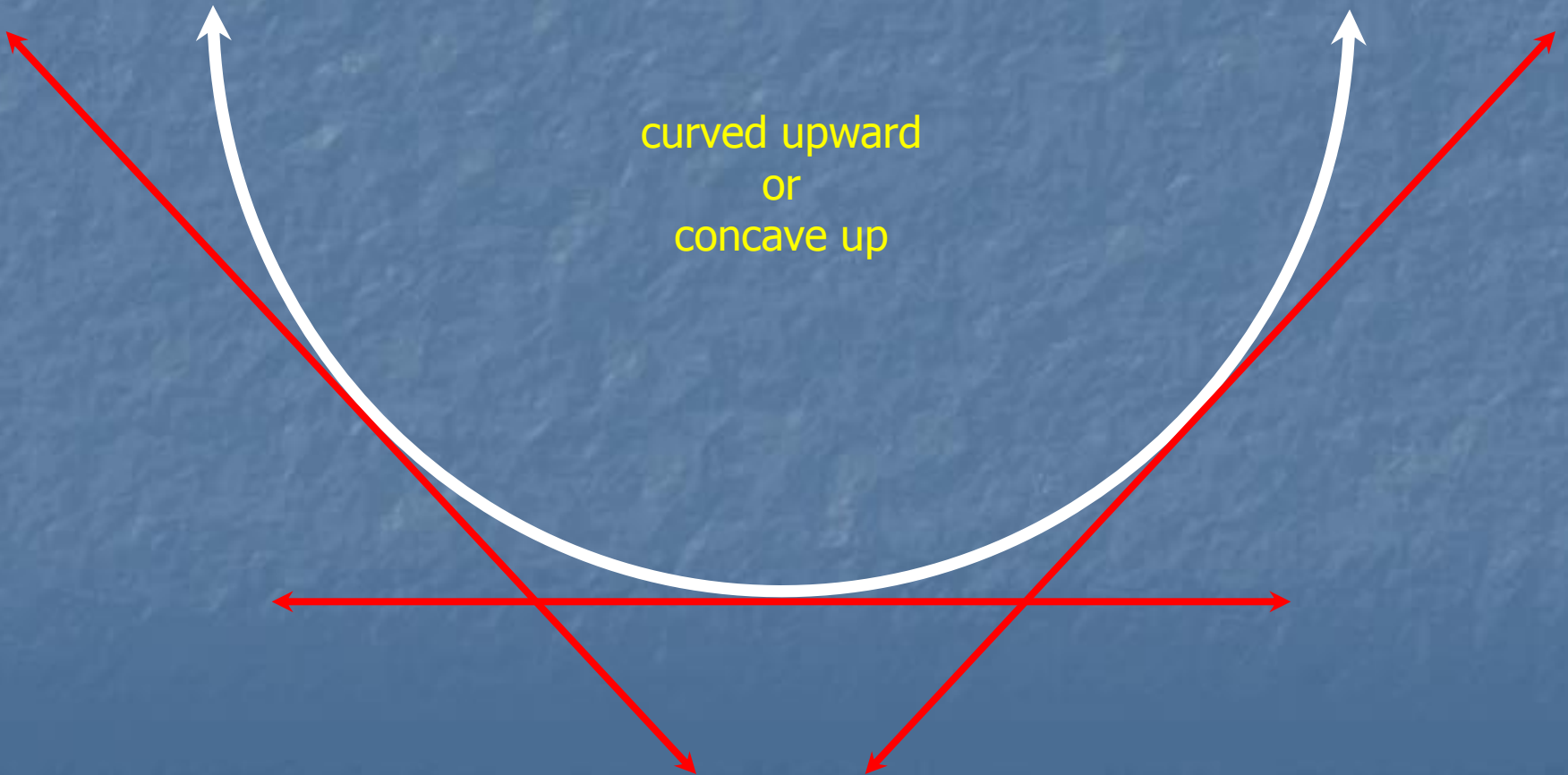
curved upward
or
concave up

Concavity



Concavity

curved upward
or
concave up

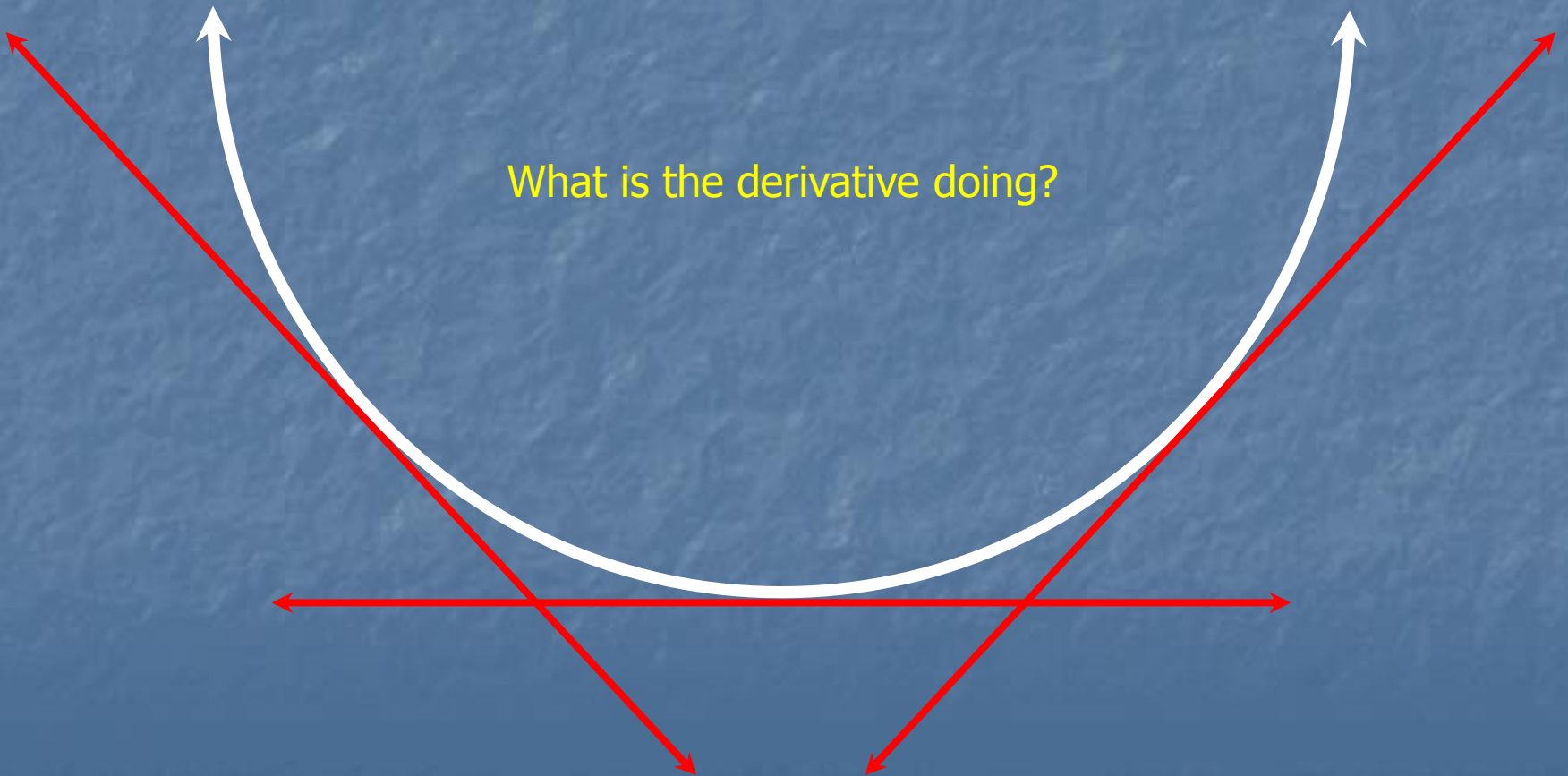


Concavity

- **Question:** Is the slope of the tangent line increasing or decreasing?

Concavity

What is the derivative doing?



Concavity

- **Question:** Is the slope of the tangent line increasing or decreasing?
- **Answer:** The slope is increasing.
- The derivative must be increasing.

Concavity

- **Question:** How do we determine where the derivative is increasing?

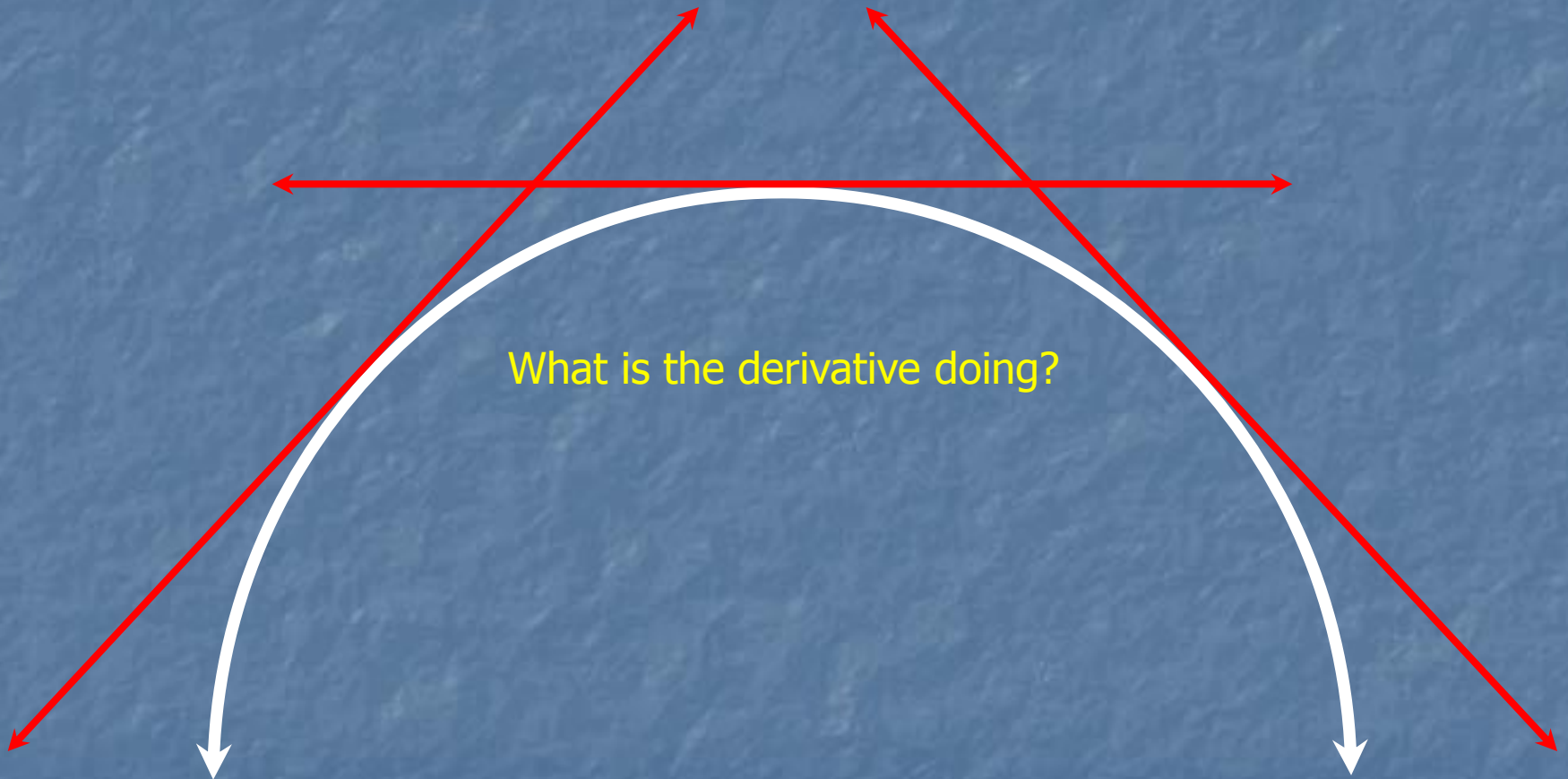
Concavity

- **Question:** How do we determine where a function is increasing?
- $f(x)$ is increasing if $f'(x) > 0$.

Concavity

- **Question:** How do we determine where the derivative is increasing?
- $f''(x)$ is increasing if $f''(x) > 0$.
- **Answer:** We must find where the second derivative is positive.

Concavity



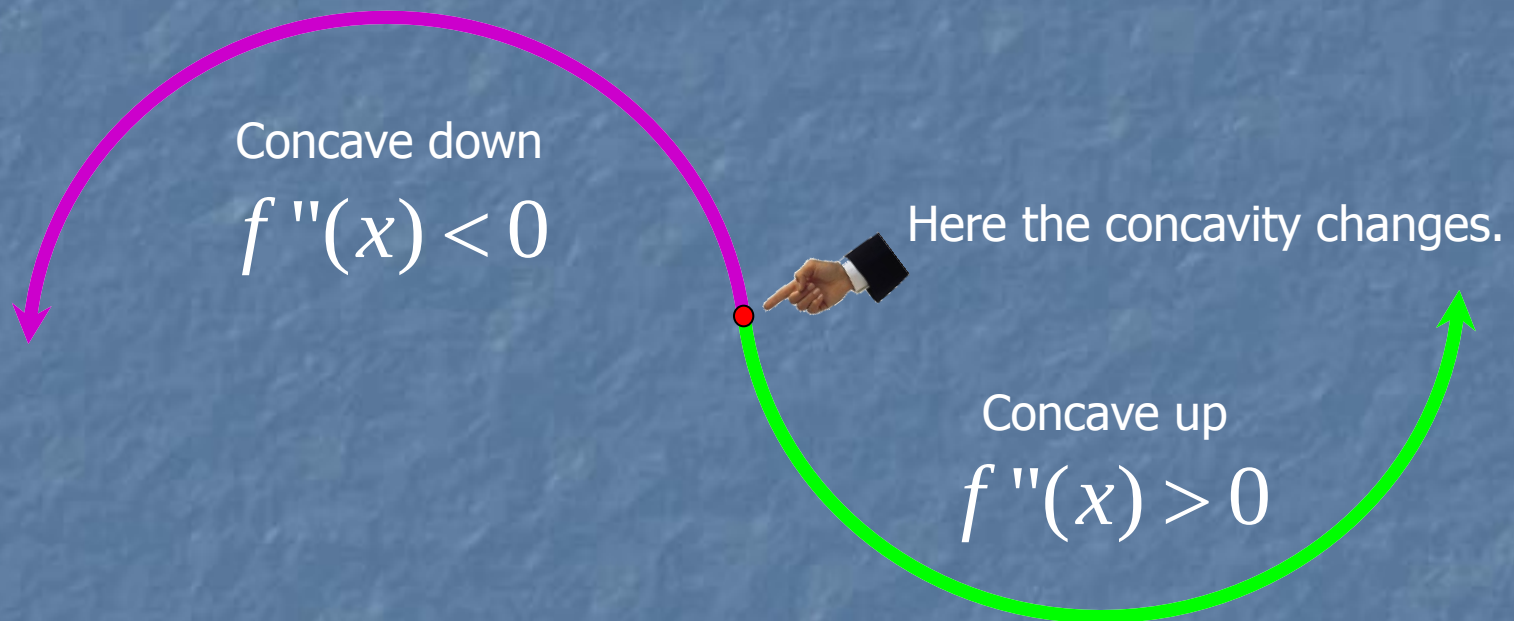
Concavity

- The concavity of a graph can be determined by using the second derivative.
- If the second derivative of a function is positive on a given interval, then the graph of the function is concave up on that interval.
- If the second derivative of a function is negative on a given interval, then the graph of the function is concave down on that interval.

The Second Derivative

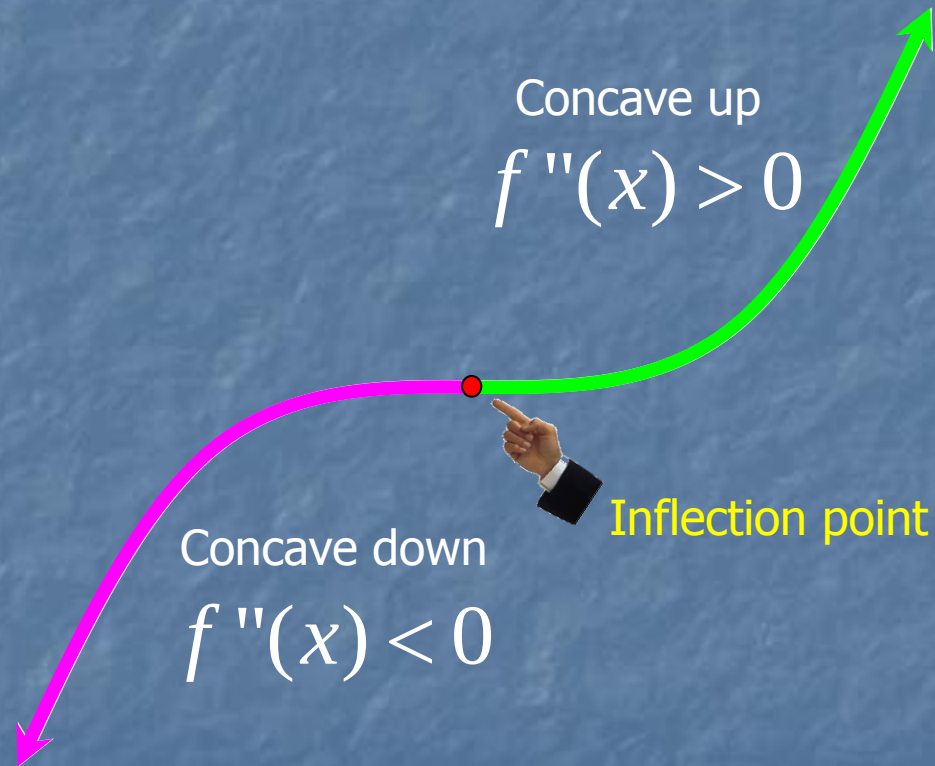
- If $f''(x) > 0$, then $f(x)$ is concave up.
- If $f''(x) < 0$, then $f(x)$ is concave down.

Concavity



This is called an **inflection point** (or point of inflection).

Concavity



Inflection Points

- **Inflection points** are points where the graph changes concavity.
- The second derivative will either equal zero or be undefined at an inflection point.

Concavity

- Find the intervals on which the function is concave up or concave down and the coordinates of any inflection points:

$$f(x) = 4x^2 + 16x - 2$$

$$f'(x) = 8x + 16$$

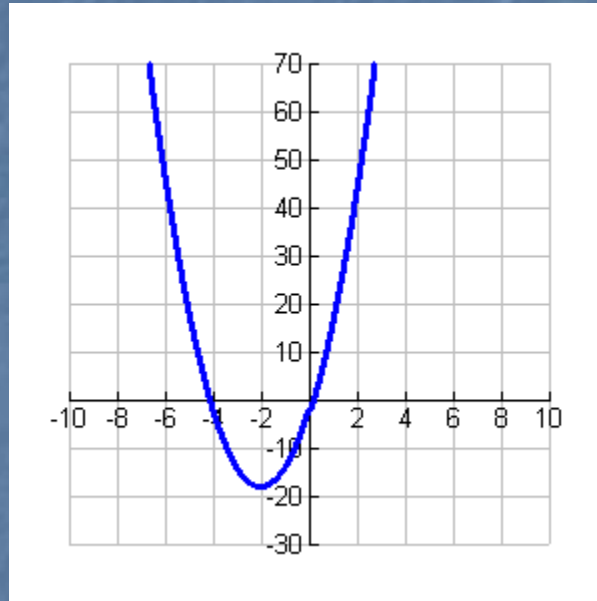
$$f''(x) = 8$$

$$f''(x) > 0$$

Always Concave up

Concavity

$$f(x) = 4x^2 + 16x - 2$$



Concave up: $(-\infty, \infty)$

Concave down: Never

Concavity

- Find the intervals on which the function is concave up or concave down and the coordinates of any inflection points:

$$g(x) = x^3 - 3x^2 - 9x + 1$$

$$g'(x) = 3x^2 - 6x - 9$$

$$g''(x) = 6x - 6$$

$$0 = 6(x - 1)$$

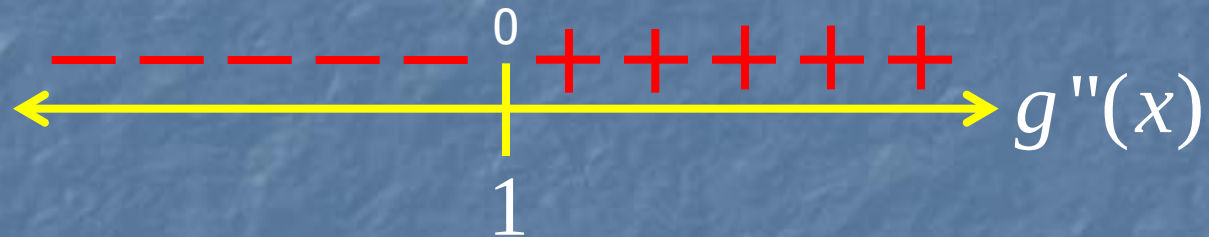
$$x - 1 = 0$$

$$x = 1$$

Concavity

$$g(x) = x^3 - 3x^2 - 9x + 1$$

$$g''(x) = 6x - 6$$



$$x = 0$$

$$x = 2$$

$$g''(0) < 0$$

$$g''(2) > 0$$

Concave down: $(-\infty, 1)$

Concave up: $(1, \infty)$

Inflection Point

$$g(x) = x^3 - 3x^2 - 9x + 1$$

$$g(1) = (1)^3 - 3(1)^2 - 9(1) + 1$$

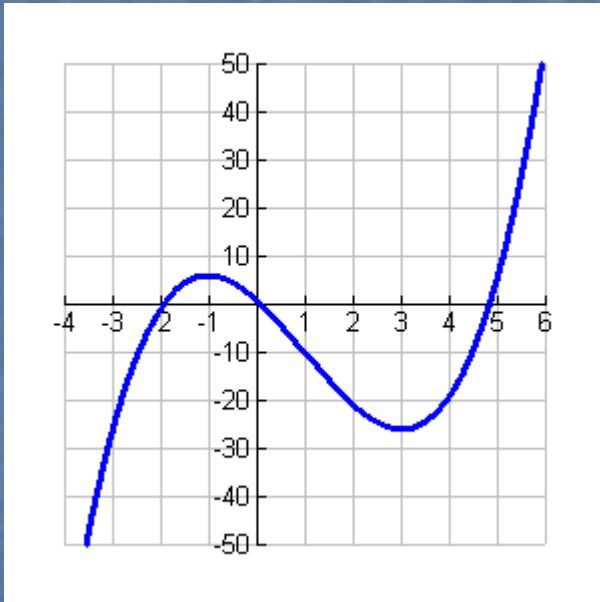
$$g(1) = 1 - 3 - 9 + 1$$

$$g(1) = -10$$

Inflection Point: (1, -10)

Concavity

$$g(x) = x^3 - 3x^2 - 9x + 1$$



Concave down: $(-\infty, 1)$

Concave up: $(1, \infty)$

Inflection Point: $(1, -10)$

Concavity

- Find the intervals on which the function is concave up or concave down and the coordinates of any inflection points:

$$h(x) = x^{1/3}$$

$$h'(x) = \frac{1}{3} x^{-2/3}$$

$$h''(x) = -\frac{2}{9} x^{-5/3}$$

$$h''(x) = -\frac{2}{9\sqrt[3]{x^5}}$$

Concavity

$$h(x) = x^{1/3}$$

$$h''(x) = -\frac{2}{9\sqrt[3]{x^5}}$$



$$x = -1$$

$$h''(-1) > 0$$

$$x = 1$$

$$h''(1) < 0$$

Concave up: $(-\infty, 0)$

Concave down: $(0, \infty)$

Inflection Point

$$h(x) = x^{1/3}$$

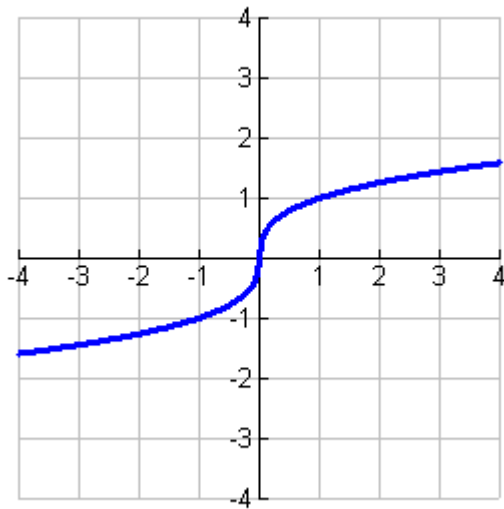
$$h(0) = (0)^{1/3}$$

$$h(0) = 0$$

Inflection Point: (0, 0)

Concavity

$$h(x) = x^{1/3}$$



Concave up: $(-\infty, 0)$

Concave down: $(0, \infty)$

Inflection Point: $(0, 0)$

Conclusion

- The second derivative can be used to determine where the graph of a function is concave up or concave down and to find inflection points.
- Knowing the critical points, increasing and decreasing intervals, relative extreme values, the concavity, and the inflection points of a function enables you to sketch accurate graphs of that function.